R15

Code No: 123AN

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, October - 2020
PROBABILITY AND STATISTICS
(Common to ME, CSE, IT, MCT, AME, MIE)

Time: 2 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) The probabilities of a Poisson variate X taking the values 1 and 2 are equal. Find:
i) Mean μ ii) $P(X \geq 1)$ iii) $P(1 < X < 4)$.
b) Given that the mean heights of students in a class is 158 cms with a standard deviation of 20 cms. Find how many students height lie between 150 cms and 170 cms. [7+8]

2.a) Average number of accidents on any day on a national highway is 1.6. Determine the probability that the number of accidents is:
i) At least one ii) At the most one.
b) The marks obtained by 500 students is normally distributed with mean 65% and Standard deviation 8%. Determine how many get more than 80%. [7+8]

3. Calculate the coefficient of rank correlation from the following data. [15]

x	68	64	75	50	64	80	75	40	55	64
y	62	58	68	45	81	60	68	48	50	70

4. The joint probability density function is given by

$$f(x) = \begin{cases} x+y, 0 < x < 1, 0 < y < 1 \\ 0, elsewhere \end{cases}$$

a) Marginal probability density function for X
b) Marginal probability density function for Y. [15]

5. A survey of 400 families with 5 children each revealed the following distribution. Is this result with the hypothesis that male births are equally probable? [15]

No. of boys	0	1	2	3	4	5
No. of families	25	60	110	113	62	30

6.a) The mean height of 90 students in a class is 180 cms Test at 90% level whether the sample had been drawn from a population whose mean is 170 cms and standard deviation 35.
b) Among the items produced by a factory out of 500, 15 were defective in another sample out of 400, 20 were defective. Test the significance between the difference of two proportions at 5% level. [7+8]

7.a) Derive the formula for the waiting time of a customer has to wait before getting service.
b) Derive formula for Expected number of customers in the system. [7+8]

8. There are three brands of tooth pastes, 1, 2 and 3. Transition of customers moving from one brand to another brand is given by the following transition probability matrix $\begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.05 & 0.75 & 0.2 \\ 0.4 & 0.3 & 0.3 \end{bmatrix}$. What are the long run market shares? [15]

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